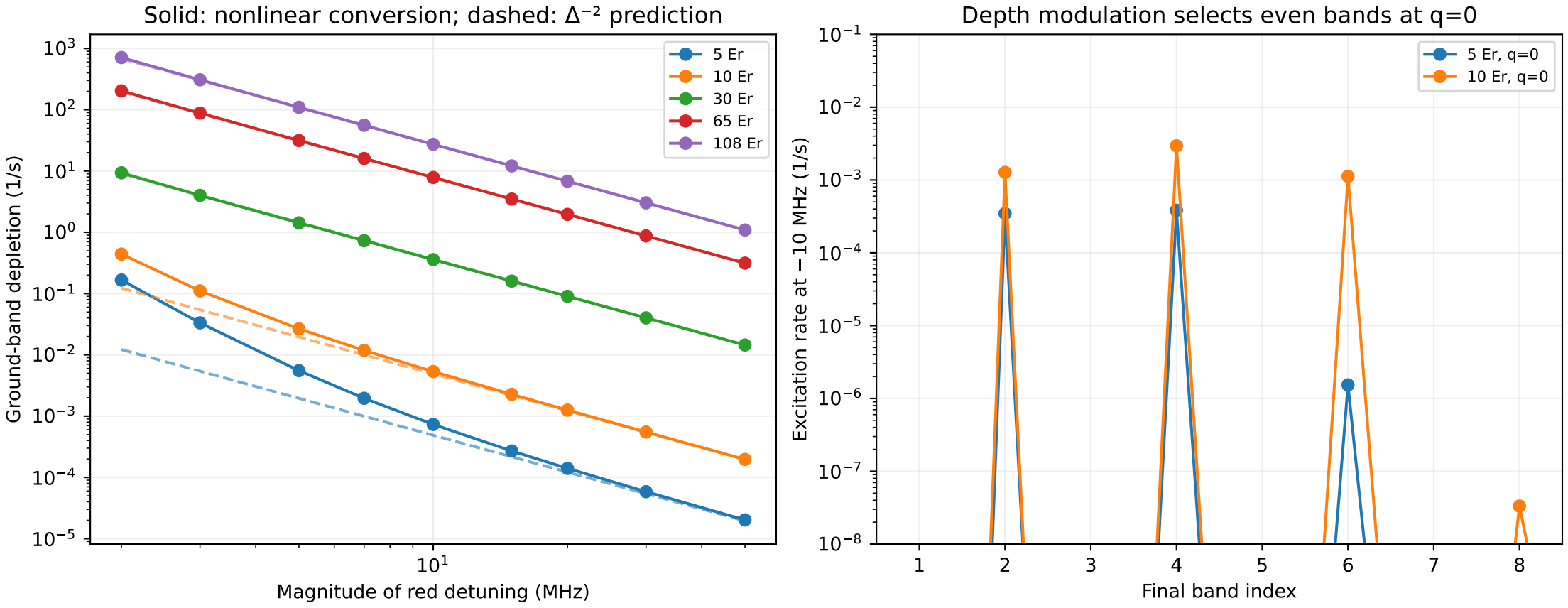

Nonlinearity mixes frequency noise down into weak low-frequency regions.
For a broad quasimomentum population, average matrix-element-weighted rates over q; see numerical summary.